O'ZBEKISTON RESPUBLIKASI RAQAMLI TEXNOLOGIYALARI VAZIRLIGI
MUHAMMAD AL-XORAZMIY NOMIDAGI TOSHKENT AXBOROT TEXNOLOGIYALARI UNIVERSITETI
FAKULTET: DASTURIY INJINIRING



Topshiriq №1

Guruh: SRM 001-3
Bajardi: Nishanov Abubakir
Tekshirdi: Bekimmetova Durdona

Toshkent – 2026

12	N ta talabaning bahosidan (0–100 oralig'ida) o'rtacha, eng yuqori, eng past bahoni va o'rtachadan yuqori talabalar sonini toping.	void statistika(const int *a, int n, double *ort, int *mx, int *mn) funksiyasini yozing.

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

int statistika(const int *a, int n, double *ort, int *mx, int *mn) {
    if (n <= 0) {
        *ort = 0;
        *mx = 0;
        *mn = 0;
        return 0;
    }

    int sum = 0;
    *mx = a[0];
    *mn = a[0];

    for (int i = 0; i < n; i++) {
        sum += a[i];
        if (a[i] > *mx) {
            *mx = a[i];
        }
        if (a[i] < *mn) {
            *mn = a[i];
        }
    }

    *ort = (double)sum / n;

    int yuqori_soni = 0;
    for (int i = 0; i < n; i++) {
        if (a[i] > *ort) {
            yuqori_soni++;
        }
    }

    return yuqori_soni;
}

int main() {
    int n;

    printf("Talabalar sonini kiriting (N): ");
    if (scanf("%d", &n) != 1 || n <= 0) {
```

```
        printf("Noto'g'ri qiymat kiritildi!\n");
        return 1;
    }

    int baholar[n];

    srand(time(NULL));

    printf("\n%d ta talabaning tasodifiy baholari (0-100 oralig'ida):\n", n);
    for (int i = 0; i < n; i++) {
        baholar[i] = rand() % 101; // 0 dan 100 gacha tasodifiy son
        printf("%d-talaba bahosi: %d\n", i + 1, baholar[i]);
    }

    double ortacha;
    int max_baho, min_baho;

    int yuqori_talabalar_soni = statistika(baholar, n, &ortacha, &max_baho, &min_baho);

    printf("\n--- NATIJALAR ---\n");
    printf("O'rtacha baho: %.2f\n", ortacha);
    printf("Eng yuqori baho: %d\n", max_baho);
    printf("Eng past baho: %d\n", min_baho);
    printf("O'rtachadan yuqori talabalar soni: %d\n", yuqori_talabalar_soni);

    return 0;
}
```

```
13-talaba bahosi: 90
14-talaba bahosi: 98
15-talaba bahosi: 65
16-talaba bahosi: 82
17-talaba bahosi: 12
18-talaba bahosi: 35
19-talaba bahosi: 23
20-talaba bahosi: 10
21-talaba bahosi: 97
22-talaba bahosi: 67
23-talaba bahosi: 99

--- NATIJALAR ---
O'rtacha baho: 60.09
Eng yuqori baho: 100
Eng past baho: 2
O'rtachadan yuqori talabalar soni: 13


...Program finished with exit code 0
Press ENTER to exit console.
```

12	Standart strdup, strcpy va strlen funksiyalarining o'z analoglarini faqat pointerlar bilan yozing.

```
#include <stdio.h>
#include <stdlib.h>

size_t my_strlen(const char *s) {
```

```
    const char *p = s;
    while (*p != '\0') {
        p++;
    }
    return (size_t)(p - s);
}

char* my_strcpy(char *dest, const char *src) {
    char *d = dest;
    while ((*d++ = *src++) != '\0');
    return dest;
}

char* my_strdup(const char *s) {

    size_t len = my_strlen(s) + 1;

    char *dest = (char*)malloc(len);
    if (dest == NULL) {
        return NULL;
    }

    my_strcpy(dest, s);

    return dest;
}

int main() {
    const char *asl_matn = "Salom, Talabalar!";

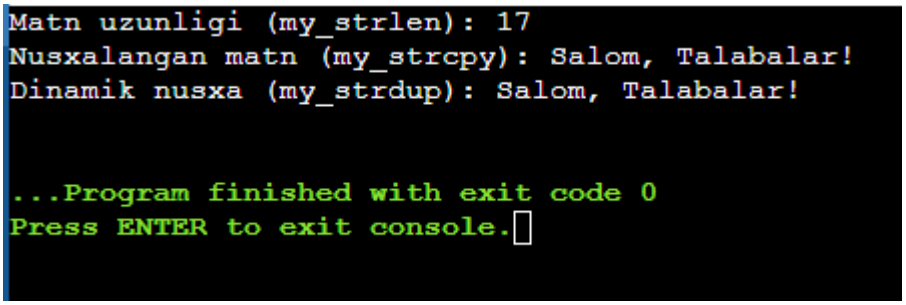
    size_t uzunlik = my_strlen(asl_matn);
    printf("Matn uzunligi (my_strlen): %zu\n", uzunlik);

    char nusxa_matn[50];
    my_strcpy(nusxa_matn, asl_matn);
    printf("Nusxalangan matn (my_strcpy): %s\n", nusxa_matn);

    char *dinamik_nusxa = my_strdup(asl_matn);
    if (dinamik_nusxa != NULL) {
        printf("Dinamik nusxa (my_strdup): %s\n", dinamik_nusxa);

        free(dinamik_nusxa);
    }

    return 0;
}
```


Matn uzunligi (my_strlen): 17
Nusxalangan matn (my_strcpy): Salom, Talabalar!
Dinamik nusxa (my_strdup): Salom, Talabalar!
...Program finished with exit code 0
Press ENTER to exit console.